

Singleton Enumeration as a Generalized Board Game

Proof Horizons, Ocean Navigation, and Password Search

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Abstract

This note develops a common language for three seemingly different problems: universal proof search, navigation through a proof Ocean, and password search. The unifying feature is *singleton enumeration*: there may be only one designated target, while the space of possible candidate histories leading to that target is enormous. Generalized board games (GBGs) provide a natural formalism for separating the cardinality of the target set from the complexity of the surrounding search space. The Universal Proof-Horizon perspective supplies the corresponding proof-search interpretation: one may know exactly what theorem is sought while still facing a combinatorially enormous or practically prohibitive space of candidate derivations. The central distinction developed here is between *structured singleton search*, where observable local features correlate with progress toward the target, and *opaque singleton search*, where no such navigational signal is available.

1 The central question

A singleton target can still induce an enormous search problem. If the target set is

$$\mathcal{T} = \{t\},$$

then

$$|\mathcal{T}| = 1,$$

but this says almost nothing about the number of legal states, legal moves, or candidate histories that must be considered before reaching t .

This observation is important for automated theorem proving. A theorem prover may be given exactly one conjecture. It is not choosing among millions of possible goals. Nevertheless, it may have to distinguish among an enormous number of possible inference paths, intermediate states, lemmas, branches, and dead ends.

The same basic shape occurs in password search. There is one accepted password, but there may be an astronomical number of possible candidates.

Thus the relevant difficulty is not necessarily the number of targets. It is the size and information structure of the space that surrounds the target.

2 Universal Proof Horizon and enumeration

The Universal Proof-Horizon viewpoint begins with a fixed effective proof system, a mechanically checkable proof relation, and a computable proof-cost measure [1]. One may enumerate candidate proofs in increasing cost and return the first valid proof of the target. When a proof exists, this

gives a canonical minimum-cost proof under the chosen presentation and cost measure. When no proof exists, the computation need not terminate.

The theoretical point is positive but the practical problem is severe: enumeration may be computationally disastrous.

The proof target can be fixed in advance. The difficulty lies in the size of the candidate proof space:

$$\{\pi : \pi \text{ is a syntactically admissible candidate proof object}\}.$$

This leads naturally to a singleton-search interpretation. For a fixed theorem T , the prover seeks one acceptable certificate:

$$\mathcal{W}_T = \{\pi : \pi \text{ verifies as a proof of } T\}.$$

In the most extreme formulation one may select a canonical proof, in which case the designated target certificate itself can be treated as a singleton.

The important distinction is therefore

$$\text{one desired outcome} \not\approx \text{small search problem}.$$

3 The Ocean as a one-player generalized board game

Consider a finite directed graph

$$G = (V, E)$$

with distinguished source s and target t . An edge

$$u \rightarrow v$$

represents one legal inference step.

This can be interpreted as a one-player generalized board game in the sense of the GBG framework [2].

Definition 1 (Ocean GBG). *The Ocean generalized board game has:*

- *game states* V ;
- *initial state* s ;
- *legal moves given by the directed edges* E ;
- *winning set*

$$\mathcal{W} = \{t\}.$$

A play is a directed path beginning at s , and a winning play is a path that reaches t .

Under unit edge cost, the minimum winning length is

$$L^* = d_G(s, t),$$

the shortest directed-path distance from s to t .

Thus the proof horizon and the game-theoretic winning horizon coincide:

$L^* = \text{minimum number of legal moves needed to reach the unique winning state.}$

The fact that the winning set has one element does not make the game trivial. The directed graph may contain a very large number of states and an even larger number of legal histories.

4 The search process as a higher-level game

A theorem prover does not merely occupy one graph vertex. Its actual state contains information about what it has already discovered.

A more faithful search state may be represented schematically as

$$S = (D, F, X, P, \Sigma, \dots),$$

where

- D is the set of discovered states;
- F is the current frontier;
- X is the set of expanded states;
- P is a predecessor or certificate map;
- Σ contains heuristic scores, reconnaissance data, or other navigational information.

A search move is then something such as

$$\text{Expand}(v), \quad v \in F.$$

This creates a second GBG: the board is no longer merely the Ocean graph but the prover's evolving knowledge of the Ocean.

In this interpretation, a system such as DATA MIND is playing a game whose moves determine which part of the underlying proof game becomes visible next.

5 Password space as a generalized board game

Password space can also be formalized as a GBG in more than one way.

5.1 Prefix-construction game

Let $\mathcal{A}$ be a finite alphabet with

$$|\mathcal{A}| = k,$$

and suppose passwords have fixed length m . Let the hidden accepted password be

$$w^* \in \mathcal{A}^m.$$

The initial state is the empty prefix:

$$s_0 = \epsilon.$$

For every symbol $a \in \mathcal{A}$, define the legal move

$$r_a(u) = ua$$

whenever

$$|u| < m.$$

The state space is

$$\mathcal{S} = \mathcal{A}^{\leq m},$$

and the winning set is

$$\mathcal{W} = \{w^*\}.$$

The resulting state graph is a rooted k -ary tree. There are

$$k^m$$

terminal strings of length m , although exactly one is winning.

Define

$$d(u) = m - |u|.$$

Then every legal character-extension move satisfies

$$d(r_a(u)) = d(u) - 1.$$

Thus the game is finite, acyclic, and has a single winning leaf.

5.2 Password-testing game and the observation map

For search, the full game state must remain rule-complete: it must contain the hidden password even though the searcher is not allowed to observe it. Let

$$\Omega = \mathcal{A}^m$$

be the set of all candidate passwords, and let the hidden accepted password be $W \in \Omega$. A full state is

$$x = (W, G, U, R),$$

where

- $G \subseteq \Omega$ is the set of candidates already guessed;
- $U = \Omega \setminus G$ is the set of untested candidates; and
- R is the transcript of responses obtained so far.

The initial full state associated with W is

$$x_0(W) = (W, \emptyset, \Omega, \emptyset).$$

A legal move is

$$\text{Guess}(q), \quad q \in U.$$

Because W is part of the full state, the transition is deterministic. If $q = W$, the successor is a terminal winning state with the response “correct.” Otherwise the successor satisfies

$$G' = G \cup \{q\}, \quad U' = U \setminus \{q\},$$

and appends “incorrect” to the transcript.

The searcher does not observe the complete state. Instead define the observation map

$$O(W, G, U, R) = (G, U, R).$$

Thus the underlying GBG remains a deterministic rule-complete transition system, while the password seeker acts under partial observation. This separation is the formal version of the

distinction between the board itself and what the player is permitted to know about the board. It is also the right setting for comparing password search with resource-sensitive settlement geometry and proof-search navigation [2, 3].

This formulation is structurally close to ATP search:

$$\text{Guess}(q) \leftrightarrow \text{Expand}(v).$$

In both cases an agent must decide which candidate to inspect next, but the information available for making that choice may differ dramatically.

5.3 Observation-constrained settlement geometry

The observation map allows intrinsic state geometry to be separated from the information geometry available to an executable policy.

Definition 2 (Observation-admissible policy). *Let X be a deterministic settlement or GBG state space and let $O : X \rightarrow \mathcal{O}$ be an observation map. A policy is O -admissible if its action at each stage depends only on its own randomness and on the history of observations and actions, not on hidden coordinates of the full state.*

Definition 3 (Observation-constrained expected settlement value). *Let μ be a probability distribution over possible initial full states, and let C_π denote the total settlement cost incurred by a policy π . Define*

$$V_{\text{Bayes}}^{O,\mu} := \inf_{\pi \in \Pi_O} \mathbb{E}_{X_0 \sim \mu, \pi} [C_\pi(X_0)],$$

where Π_O is the class of O -admissible policies. With full observation $O = \text{id}$, the corresponding value is denoted $V_{\text{Bayes}}^{\text{id},\mu}$.

This quantity is intentionally distinct from the intrinsic Bellman value $V(x)$ of a fully specified state. The intrinsic value measures the cheapest settling trajectory that exists from x ; the observation-constrained value measures the best expected cost attainable when the policy cannot see all of x . In a password game these quantities can differ exponentially. In a proof Ocean, by contrast, local graph structure may expose navigational signal that substantially narrows the gap.

6 Why password search and Ocean search are not identical

The password analogy becomes most useful when its limitation is made explicit.

Suppose an ideal password oracle returns only

correct or incorrect

after a complete candidate is submitted. Assume further that the hidden password is uniformly distributed and that no side-channel information is available.

Then, before testing, the untried candidates are symmetric. A failed guess does not reveal that a nearby candidate is more promising.

There is no useful local geometry.

By contrast, a structured proof Ocean can contain local information. Some branches may terminate quickly, some may continue for many levels, some may have distinctive branching patterns, and some local configurations may correlate with eventual reachability of the target.

A navigation policy can exploit such differences.

This gives the following comparison.

	Structured Ocean search	Ideal password search
Unique target	Yes	Yes
Large candidate space	Yes	Yes
Branching	Yes	Yes
Observable local structure varies	Often	No, ideally
Dead-end or continuation clues	Possible	Absent
Navigation can exploit local signal	Yes	Not without extra information

7 Structured and opaque singleton search

The preceding comparison suggests a useful distinction.

Definition 4 (Structured singleton search). *Relative to an observation map O , a singleton search problem is structured when observable features of nonterminal states contain information correlated with accessibility, distance, or probability of eventual success at the unique target.*

Definition 5 (Opaque singleton search). *Relative to an observation map O , a singleton search problem is opaque when states or untested candidates that remain possible under the current observation history are indistinguishable with respect to all information available to the searcher, except through direct testing.*

The Ocean is intended to model the first situation. Ideal password guessing models the second.

Principle 1 (Singleton Navigation Principle). *A singleton target does not prevent intelligent navigation. What matters is whether the observable search state contains information correlated with progress toward the unique winning state.*

This principle separates two questions that are easy to conflate:

How many targets are there?

and

How informative is the surrounding search space?

The first may equal one while the second remains highly nontrivial.

8 DATA MIND in the GBG interpretation

Under this framework, DATA MIND is not successful merely because it can “enumerate toward one answer.” Its potential advantage arises when the game contains navigational signal.

Its task can be written abstractly as:

choose $v \in F$ so as to maximize expected progress toward t .

A heuristic value function may be written

$V(v) \approx$ estimated usefulness of expanding v for eventually reaching t .

The essential role of such a value is to change the *order of exploration*. It does not change the legal moves, the target, or the verification standard.

Thus:

heuristic navigation $\longrightarrow$ candidate ordering,

while

formal verifier $\longrightarrow$ certificate acceptance.

This is compatible with the Universal Proof-Horizon viewpoint. The exhaustive procedure supplies a universal baseline; the navigation policy attempts to avoid examining almost all irrelevant candidates.

9 A hierarchy of singleton-search regimes

The discussion suggests a useful spectrum:

opaque guessing $\longrightarrow$ weakly structured navigation
 $\longrightarrow$ strongly structured navigation $\longrightarrow$ exact shortest-path knowledge

At the left extreme, intelligent reordering cannot outperform symmetry except through prior information or side channels.

At the right extreme, an exact distance-to-target function

$$h(v) = d(v, t)$$

makes navigation trivial: always choose a successor whose distance decreases by one.

Most interesting ATP systems lie between these extremes. They do not know the true proof horizon in advance, but they try to estimate which legal actions reduce the hidden remaining distance.

10 Connection to proof horizons

For a unit-cost finite proof graph, the true distance

$$h^*(v) = d_G(v, t)$$

is a proof-horizon function.

At the initial state,

$$h^*(s) = L^*.$$

An ideal navigator would always select a successor u satisfying

$$h^*(u) = h^*(v) - 1.$$

A learned or heuristic navigator instead uses an approximation

$$\hat{h}(v) \approx h^*(v).$$

The resulting ATP problem can therefore be viewed as a game of repeatedly estimating the hidden proof horizon while preserving exact verification at the end.

This yields the synthesis:

Universal Proof Horizon + Generalized Board Games + singleton enumeration + navigation

as a common framework for understanding proof search.

11 What the password analogy teaches

The password model is not merely an analogy to proof search. It is a useful control case.

If an ATP benchmark were completely opaque in the same sense as ideal password guessing, then no generic navigation method could extract useful directional information from local structure because no such structure would be present.

If a method performs substantially better than indiscriminate enumeration on a singleton benchmark, one should therefore ask:

1. What observable structure was available?
2. Which features of that structure correlated with eventual success?
3. Did the method exploit legitimate local signal or hidden benchmark metadata?
4. How much candidate work was avoided?
5. How close was the resulting certificate to the true proof horizon?

These questions separate genuine navigation from accidental leakage.

12 Cross-domain theorem additions

Provenance note. The bracketed Atlas note attached to each theorem below records that Atlas added and proved the statement in this manuscript. It is a manuscript-provenance statement, not a claim that every underlying mathematical fact is historically new. Several results are elementary consequences of standard search arguments or of the definitions developed in the cited Tenneson frameworks.

The preceding framework suggests several statements that can be proved without assuming that a singleton target is easy to locate. The results in this section make precise the distinction between the size of a target set, the intrinsic distance to that target, the information visible to a searcher, and the work actually spent locating a certificate. They also provide explicit bridges among password seeking, CNF/3-CNF, generalized board games, the Universal Proof-Horizon viewpoint, DATA MIND-style navigation, and fair settlement geometry.

12.1 Information-constrained Bellman gap for opaque passwords

The ordinary Bellman value of a fully specified state assumes that the state contains whatever information is needed to choose among legal actions. A password searcher generally does not observe the hidden password itself. This creates a useful distinction between a full-information Bellman value and an observation-constrained value.

Let Ω be a set of $N \geq 1$ candidate passwords, let W be uniformly distributed on Ω , and use the full state and observation map defined above. A query $q \in \Omega$ receives only the answer “correct” when $q = W$ and “incorrect” otherwise. An admissible search policy is required to be O -admissible, nonrepeating on failed queries, and to continue until success. Let T_π be its number of guesses. For every full state the intrinsic full-information Bellman cost is one guess.

Theorem 12.1 (Information-Constrained Bellman Gap). *For the equality-only password game above, every admissible possibly randomized nonrepeating policy π satisfies, for $0 \leq k \leq N$,*

$$\Pr(T_\pi \leq k) = \frac{k}{N}$$

and

$$\mathbb{E}[T_\pi] = \frac{N+1}{2}.$$

If

$$V_{\text{Bayes}}^{\text{id},\mu} = 1$$

is the full-observation expected settlement value and

$$V_{\text{Bayes}}^{O,\mu} := \inf_{\pi \in \Pi_O} \mathbb{E}[T_\pi],$$

then

$$\boxed{\frac{V_{\text{Bayes}}^{O,\mu}}{V_{\text{Bayes}}^{\text{id},\mu}} = \frac{N+1}{2}}.$$

In particular, for $N = 2^n$ the information-constrained Bellman gap is exponential in n .

[Added and proved by Atlas, a ChatGPT ATP/AMLD add/on designed by Brian Tenneson]

Proof. Condition first on the internal randomness of π . Until the successful query occurs, every observed answer is “incorrect.” Consequently a nonrepeating deterministic realization of the policy specifies an ordering

$$q_1, q_2, \dots, q_N$$

of the candidates. Since W is uniform on Ω , its rank in this ordering is uniform on $\{1, \dots, N\}$. Hence

$$\Pr(T_\pi \leq k) = \frac{k}{N}$$

and

$$\mathbb{E}[T_\pi] = \frac{1}{N} \sum_{j=1}^N j = \frac{N+1}{2}.$$

These identities hold for every fixed realization of the policy’s randomness, so averaging over that randomness leaves them unchanged. A full-information controller guesses W immediately, giving $V_{\text{Bayes}}^{\text{id},\mu} = 1$. The stated ratio follows. $\square$

Discussion. This theorem formalizes the sense in which an ideal password problem is an *opaque* singleton GBG. The winning set has one element, but the searcher is not allowed to observe the coordinate that identifies that element. The theorem therefore separates geometric accessibility from informational accessibility. In the full state space the winning state is one legal action away; under the observation map available to the searcher, the expected number of tests is $(N+1)/2$.

This suggests an extension of settlement geometry in which a declared observation map $O : X \rightarrow \mathcal{O}$ restricts the policies that may act on a state. The intrinsic full-state Bellman value and the observation-constrained expected value $V_{\text{Bayes}}^{O,\mu}$ can then differ enormously. This is precisely the kind of distinction needed when comparing a structurally informative proof Ocean with an opaque password oracle. It also explains why a short route existing in a state graph does not by itself imply that a searcher can identify that route cheaply.

12.2 Opaque passwords cannot settle with high probability in polynomially many guesses

The first theorem immediately gives an asymptotic statement. If the password has n uniformly random bits, the candidate set has size 2^n . A solver that is exact—that is, it never reports success without receiving the positive oracle response—may use a bounded number of guesses and otherwise return UNKNOWN. This is the equality-query analogue of verifier-gated high-probability settlement: correctness is exact, while completion is resource bounded. The observation-constrained model is an extension of, rather than an identification with, the fully visible settlement systems of Fair Settlement Geometry [3].

Theorem 12.2 (Opaque Singleton No-Polynomial-Settlement). *Let W be a uniformly random n -bit password and suppose the only feedback from a guess is equality with W . If an exact target-blind policy makes at most $q(n)$ distinct guesses, then*

$$\Pr(\text{settlement within } q(n) \text{ guesses}) \leq \min \left\{ 1, \frac{q(n)}{2^n} \right\}.$$

Equality holds when the policy is prepared to test exactly $\min\{q(n), 2^n\}$ distinct candidates unless success occurs earlier. Consequently, for every polynomial q ,

$$\Pr(\text{settlement within } q(n) \text{ guesses}) \longrightarrow 0.$$

Thus, in the observation-constrained equality-query model, the uniform opaque-password family has no high-probability polynomial settlement policy.

[Added and proved by Atlas, a ChatGPT ATP/AMLD add/on designed by Brian Tenneson]

Proof. A policy using at most $q(n)$ guesses tests at most $\min\{q(n), 2^n\}$ distinct members of the candidate set. Conditional on any fixed realization of the policy’s internal randomness, those tested candidates occupy at most that many positions in the induced ordering. Since W is uniform, the probability that its rank lies among them is at most

$$\min \left\{ 1, \frac{q(n)}{2^n} \right\}.$$

If the policy continues through exactly that many distinct candidates unless success occurs earlier, the bound is attained. If q is polynomial, then $q(n)/2^n \rightarrow 0$. $\square$

Discussion. This result supplies a zero-navigation-information control for the broader program. A system can be perfectly sound, can have an unambiguous singleton target, and can even possess a one-step full-information route to success, yet an observation-limited searcher still fails with probability tending to one under every polynomial query budget.

The relevance to DATA MIND is negative but important: if a benchmark were truly password-like in this equality-only sense, no amount of clever local navigation could create information that is absent from the interface. Therefore strong performance on a singleton benchmark is evidence either that useful structure is observable, that informative prior knowledge is available, or that the benchmark leaks information. A contamination-safe experiment must distinguish these possibilities.

12.3 Constraint revelation in CNF and 3-CNF

Password space becomes more interesting when it is compared with SAT. A known CNF formula exposes constraints before a complete assignment is tested. Partial assignments can already falsify clauses, and those clauses prune the prefix tree. The next theorem quantifies this effect. Although motivated by 3-CNF, the proof works for arbitrary CNF.

Fix a variable ordering $x_1, \dots, x_n$ and let φ be a satisfiable CNF formula. Let $P_k(\varphi)$ be the set of assignments to $x_1, \dots, x_k$ that have not yet falsified any clause of φ , and write

$$W_k := |P_k(\varphi)|.$$

Define the prefix-pruning, or constraint-revelation, index

$$J_k := \log_2 \left(\frac{2^k}{W_k} \right) = k - \log_2 W_k.$$

Theorem 12.3 (CNF Prefix-Pruning Monotonicity). *If φ is a satisfiable CNF formula on $x_1, \dots, x_n$, then*

$$W_0 = 1, \quad 1 \leq W_k \leq 2^k, \quad W_{k+1} \leq 2W_k,$$

and

$$W_n = \# \text{SAT}(\varphi).$$

Consequently,

$$\boxed{0 = J_0 \leq J_1 \leq \dots \leq J_n = n - \log_2 \# \text{SAT}(\varphi).}$$

In the uniquely satisfiable case, $W_n = 1$ and hence $J_n = n$. The same statement applies in particular to satisfiable and uniquely satisfiable 3-CNF formulas.

[Added and proved by Atlas, a ChatGPT ATP/AMLD add/on designed by Brian Tenneson]

Proof. The empty assignment is the unique length-zero prefix, so $W_0 = 1$. The restriction of any satisfying assignment to the first k variables cannot falsify a clause; hence satisfiability gives $W_k \geq 1$. Trivially $W_k \leq 2^k$.

Every surviving k -prefix has at most two one-variable extensions, obtained by setting x_{k+1} to 0 or 1. Some extensions may falsify a clause, so

$$W_{k+1} \leq 2W_k.$$

At depth n , a complete assignment fails to falsify any clause exactly when it satisfies the CNF formula. Therefore

$$W_n = \# \text{SAT}(\varphi).$$

Finally,

$$J_{k+1} - J_k = 1 - \log_2 \left(\frac{W_{k+1}}{W_k} \right) \geq 0,$$

because $W_{k+1}/W_k \leq 2$. Also $J_0 = 0$ and

$$J_n = n - \log_2 \# \text{SAT}(\varphi).$$

If the satisfying assignment is unique, this endpoint is n . □

Discussion. The quantity J_k provides a concrete bridge between password seeking and SAT/3-CNF. In an equality-only password problem, no prefix is rejected merely because of its internal symbols: before complete candidates are individually tested, the observable prefix tree remains symmetric. A CNF formula, by contrast, exposes local logical constraints that can eliminate prefixes long before depth n .

The theorem does *not* say that unique 3-SAT is easy, nor does monotone pruning imply that a polynomial-work navigation policy exists. It says something more modest and useful for the present framework: SAT can possess an observable information geometry that the ideal password oracle lacks. The index J_k measures accumulated pruning along a chosen variable order and can therefore be studied as a possible feature of a SAT GBG or as one component of a navigation heuristic.

12.4 Equivalence of four singleton horizons

The Ocean construction, generalized board games [2], the Universal Proof-Horizon viewpoint [1], and Bellman settlement geometry [3] all contain a notion of “distance to success.” Under a unit-cost one-player proof game these notions coincide exactly.

Theorem 12.4 (Singleton GBG–Proof-Horizon–Bellman Equivalence). *Let $G = (V, E)$ be an effectively generated, locally finite, directed one-player GBG with initial state s and singleton winning set $\{t\}$. Suppose each legal move has unit cost and the reachable GBG transition system is step-for-step identical to the associated sequential proof system: every legal GBG move is one admissible inference step, and every admissible inference step from a reachable proof state is represented by a legal GBG move. If t is reachable from s , let*

- L^* be the minimum winning match length;
- $h(t)$ be the minimum number of inference steps in a proof of t ;
- $V(s)$ be the Bellman settlement value with unit action costs; and
- $d_G(s, t)$ be the directed graph distance.

Then

$$\boxed{L^* = h(t) = V(s) = d_G(s, t).}$$

[Added and proved by Atlas, a ChatGPT ATP/AMLD add/on designed by Brian Tenneson]

Proof. By the step-for-step identity hypothesis, legal matches from s to t are exactly the admissible sequential proof paths from the initial proof state to the target. Therefore the minimum match length equals the minimum proof length, so $L^* = h(t)$. Both quantities are by definition the minimum number of directed edges in an s – t path, hence they equal $d_G(s, t)$. Since every legal action has cost one, the accumulated cost of any settling trajectory is exactly its number of edges. Taking the minimum over settling trajectories gives $V(s) = d_G(s, t)$. $\square$

Discussion. This theorem identifies a common invariant across four vocabularies. In the Ocean benchmark, L^* is not merely a convenient graph statistic: under the declared unit-cost proof-game interpretation it is simultaneously the shortest winning game, the shortest proof horizon, the directed distance to the target, and the optimal Bellman settlement cost.

The equality is valuable experimentally because it tells us what may be used as an objective denominator when measuring navigation. A search policy can be compared with the same L^*

whether one speaks the language of GBGs, proof horizons, or settlement geometry. What the equality does *not* determine is how much search work an executable policy must perform before discovering a route of that length or nearly that length. That is the subject of the next theorem.

12.5 Shortest certificates do not control search overhead

Definition 6 (Irrelevant expansion). *In a singleton-target directed search graph with target t , an expansion of a vertex v is called irrelevant if*

$$d_G(v, t) = \infty,$$

so no continuation from v can contribute to a certificate reaching t .

A proof can be optimal even when the search that found it was extremely wasteful. This matters directly for interpreting a DATA MIND Ocean run: proof length, expansion count, reconnaissance probes, memory, and wall time are different observables.

Theorem 12.5 (Horizon–Execution Separation). *For every pair of integers $L \geq 1$ and $M \geq 1$, there exists a finite singleton-target one-player GBG together with an exact search policy π such that*

$$L^* = V(s) = L,$$

the certificate eventually returned by π has exactly L legal moves, but π performs at least M irrelevant state expansions before completing that certificate. Hence certificate stretch

$$\frac{\text{returned certificate length}}{L^*} = 1$$

does not bound search overhead.

[Added and proved by Atlas, a ChatGPT ATP/AMLD add/on designed by Brian Tenneson]

Proof. Construct a directed graph with a source s , a unique successful path

$$s = v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_L = t,$$

and at least M additional decoy states reachable from s (or from a retained frontier) that do not lead to t . Give every edge unit cost. Then the unique shortest successful path has length L , so $L^* = V(s) = L$.

Now consider the higher-level search game whose state records the discovered frontier and whose legal search actions expand frontier vertices. Define π to expand the M decoy states first and then to expand the vertices on the successful path until t is reached. The predecessor map retained by the search reconstructs the original L -edge path, so the returned certificate is optimal even though at least M irrelevant expansions were spent before the search completed it. Since M is arbitrary, no bound on search overhead follows from certificate stretch alone. $\square$

Discussion. This theorem is the formal reason to keep certificate quality and search quality separate. The Fair Settlement Geometry framework [3] distinguishes an intrinsic best-route value $V_M(w)$ from the actual resource cost $C_{M,\pi}(w)$ incurred by a policy. Horizon–Execution Separation shows that this distinction is not cosmetic: even a policy that ultimately returns an optimal proof may incur arbitrarily large irrelevant work.

The Ocean experiment discussed in the surrounding research provides an empirical illustration. DATA MIND can be evaluated by how close its returned certificate lies to L^* , but its expansion count and bounded-reconnaissance probes must still be reported independently. A near-horizon proof is evidence of good certificate navigation; it is not, by itself, a theorem of low total search cost.

12.6 Efficient GBG isomorphisms preserve fair settlement degree

The GBG paper treats different games as formal transition systems, while Fair Settlement Geometry compares exact settlement systems through reductions that preserve answer semantics, construction cost, certificate transport, and Bellman geometry up to polynomial distortion. A sufficiently efficient GBG isomorphism therefore gives an immediate bridge between the two theories.

Theorem 12.6 (Efficient GBG Isomorphism Preserves Fair Settlement Degree). *Consider two families of finitely encoded, exact singleton-target GBGs, $\mathcal{G}$ and $\mathcal{H}$, equipped with settlement systems M_G and M_H . Let*

$$L_G := \{\langle I \rangle : \text{the designated target of } I \text{ is reachable}\}$$

and define L_H analogously. Suppose there are mutually inverse translations f and g , computable with polynomial declared primitive work, between their encoded instances and corresponding state/action representations such that:

- (i) initial states and yes/no settlement status are preserved;*
- (ii) legal moves and their transitions are preserved bijectively;*
- (iii) action costs are preserved exactly;*
- (iv) encoded sizes are polynomially related in both directions; and*
- (v) terminal certificates and verifier acceptance are transportable with polynomial declared primitive work in both directions.*

Then the associated decision/settlement pairs fairly reduce to one another, so

$$(L_G, M_G) \sim_{\text{fair}} (L_H, M_H).$$

In particular, they lie in the same fair settlement degree. Moreover, for corresponding states x and $f(x)$,

$$\boxed{V_{M_G}(x) = V_{M_H}(f(x)).}$$

[Added and proved by Atlas, a ChatGPT ATP/AMLD add/on designed by Brian Tenneson]

Proof. Because f maps legal actions and transitions bijectively while preserving action costs, every finite settling trajectory in $\mathcal{G}$ has an f -image in $\mathcal{H}$ of exactly the same accumulated cost. Applying the inverse translation g shows the converse. Taking infima over settling trajectories therefore yields

$$V_{M_G}(x) = V_{M_H}(f(x)).$$

The assumed polynomial declared primitive work for construction, polynomial size relation, answer preservation, and polynomial certificate transport satisfy the corresponding requirements of a fair settlement reduction from (L_G, M_G) to (L_H, M_H) [3]. The inverse translation gives the reverse fair reduction. Mutual fair reducibility is exactly equality of fair settlement degree. $\square$

Discussion. This theorem links generalized-board-game representation directly to the resource-sensitive preorder of Fair Settlement Geometry. It says that a mere change of notation or efficiently invertible presentation does not create a new fair settlement degree when the transition structure, costs, and certificates are genuinely preserved. Thus a proof-search problem can be recast as a GBG without changing its resource geometry, provided the recoding is efficient and faithful in the strong sense stated above.

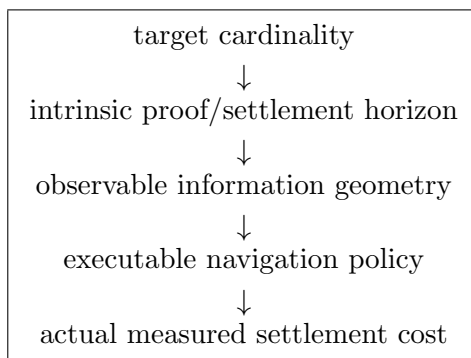
The theorem also clarifies why weaker correspondences are insufficient. A map that preserves only reachability could collapse a difficult region to an artificially cheap one, hide expensive preprocessing, or make certificate lifting costly. Fair settlement equivalence requires preservation of the computational geometry rather than merely preservation of the final yes/no answer.

12.7 Synthesis: what singleton cardinality does not determine

Taken together, the six results support a precise version of the main theme of this note. Target cardinality alone is not a complexity measure. Even after the intrinsic horizon is fixed, the difficulty experienced by a search policy can vary dramatically because information, observable constraints, and search overhead remain separate variables.

The password theorems identify the opaque extreme: a unique target can have an exponential observation-constrained search cost despite a trivial full-information distance. The CNF theorem identifies one source of structure: known clauses can destroy prefix symmetry and progressively expose constraints. The GBG–Proof-Horizon–Bellman theorem gives a common intrinsic distance for unit-cost proof games. Horizon–Execution Separation shows that this intrinsic distance still does not determine executable search work. Finally, the fair settlement theorem identifies when two different GBG presentations preserve that resource geometry.

This yields a layered view of singleton search:



No arrow is reversible in general. That non-equivalence is the conceptual space in which DATA MIND-style navigation is intended to operate.

13 Conclusion

A one-element target set can support a very difficult search problem. The difficulty belongs not to the cardinality of the target set but to the number, structure, and information content of the candidate histories surrounding it.

Password space supplies a clean GBG model of singleton search. In its ideal opaque form, it represents the limiting case in which no useful local navigation signal exists. A proof Ocean represents a contrasting case in which local graph structure may contain information correlated with eventual success.

This distinction clarifies the role of an intelligent ATP navigator. Its job is not to create new logical consequences or redefine proof. Its job is to order legal exploration so that the unique target is reached while examining as little irrelevant search space as possible.

In that sense, theorem proving can be understood as a one-player generalized board game played toward a singleton winning state, while the Universal Proof-Horizon framework [1] supplies the objective horizon against which the quality of that play can be measured.

References

- [1] Brian Tenneson, *The Universal Proof-Horizon Operator: From MIU and the n th Well-Formed Formula to ZFC, Predator, and Automated Logical Deciders*, Version 1.0, August 3, 2026.
- [2] Brian Tenneson, *Generalized Board Games: From Tic-Tac-Toe to Chess*, 2026. Available at: https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/generalized_board_games_tic_tac_toe_chess_001/.
- [3] Brian Tenneson, *Preorders and Fair Settlement Geometry: From SIC and AMLD Simulation Degrees to a High-Probability Complexity Framework for SAT*, September 2026. Available at: https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/preorders_and_fair_settlement_geometry_001/.